

F09. Integrated evidence and boundary matrix

Mechanism link	Evidence level	Scope	Boundary
physical pool size -> end to end measured behavior	MEASURED CORRELATION	CONTROLLED ONE FACTOR D4 CAPACITY SWEEP 24 32 48 KiB	This establishes workload-dependent end-to-end sensitivity to the controlled capacity change; it does not identify internal mediators.
physical pool size -> pool full time exposure	MEASURED CORRELATION	CONTROLLED ONE FACTOR D4 CAPACITY SWEEP 24 32 48 KiB	At 48 KiB pool-full time exposure falls substantially for BICG/GESUMMV, especially in OO; Btree remains a zero-full control. This is distinct from accumulated retry
physical pool size -> IO no free events per active SM cycle	MEASURED CORRELATION	CONTROLLED ONE FACTOR D4 CAPACITY SWEEP IO ONLY	IO time-exposure rate falls at 48 KiB. Whole-line OO has no emitted compact no-free counter and is not imputed.
larger pool -> cumulative no free burden per instruction	NOT SUPPORTED	CONTROLLED ONE FACTOR D4 CAPACITY SWEEP IO ONLY	The claim that a larger pool lowers accumulated no-free burden per completed instruction is not supported. Program-work burden need not follow time exposure because
physical occupancy or pool full exposure -> lower inflight concurrency	MEASURED CORRELATION	D4 INTERNAL MEDIATORS	Capacity is controlled, but neither mediator is independently intervened on; direction is not identified.
lower inflight concurrency -> L2 miss or reservation pressure	MEASURED CORRELATION	D4 INTERNAL MEDIATORS	No mediator-only intervention establishes direction or relative contribution.
L2 miss or reservation pressure -> alloc to ready pending lifetime	MEASURED CORRELATION	D4 INTERNAL MEDIATORS	The controlled sweep supports co-variation, not L2 pressure as the independently proven cause of lifetime.
alloc to ready pending lifetime -> pending tag eviction	INSUFFICIENT	D4 INTERNAL MEDIATORS	No per-allocation causal linkage or mediator intervention is measured.
pending tag eviction -> duplicate lower request traffic	SOURCE PROVEN	LANE C IO SOURCE AUDIT AND OO DUPLICATE COUNTER SEMANTICS	Exact transition semantics; it neither establishes frequency nor attributes performance.
duplicate lower request traffic -> performance	NOT SUPPORTED	D4 D5 AND ACCEPTED FAST64 SPEEDUPS	No recoverable-performance estimate or primary-limiter attribution is made.
OO pending tag eviction -> deferred physical lifetime	SOURCE PROVEN	OO DUPLICATE COUNTER SEMANTICS AND OBSERVER COUNTER SEMANTICS	Exact state/observer semantics, not a duration or performance causal claim.
OO deferred physical lifetime -> final reclaim	SOURCE PROVEN	OO DUPLICATE COUNTER SEMANTICS	Source-defined lifecycle only.
OO final reclaim -> exposed concurrency	INSUFFICIENT	D4 INTERNAL MEDIATORS	No final-reclaim intervention establishes direction.
exposed concurrency -> performance	NOT SUPPORTED	D4 INTERNAL MEDIATORS	The data do not assign performance changes to concurrency alone.
universal larger pool -> downstream pressure transfer	NOT SUPPORTED	D4 CONTROL AND WORKLOAD COMPARISON	A universal statement is contradicted; local bottleneck-migration interpretation remains evidence-bounded.
downstream L2 pressure primary limiter with duplicate feedback secondary	INSUFFICIENT	INTEGRATED D4 D5 EVIDENCE	Consistent with bottleneck migration, not proof that L2 is dominant or duplicates secondary.

Do not infer a single root cause. Source-proven, measured, insufficient, and not-supported results are distinct.